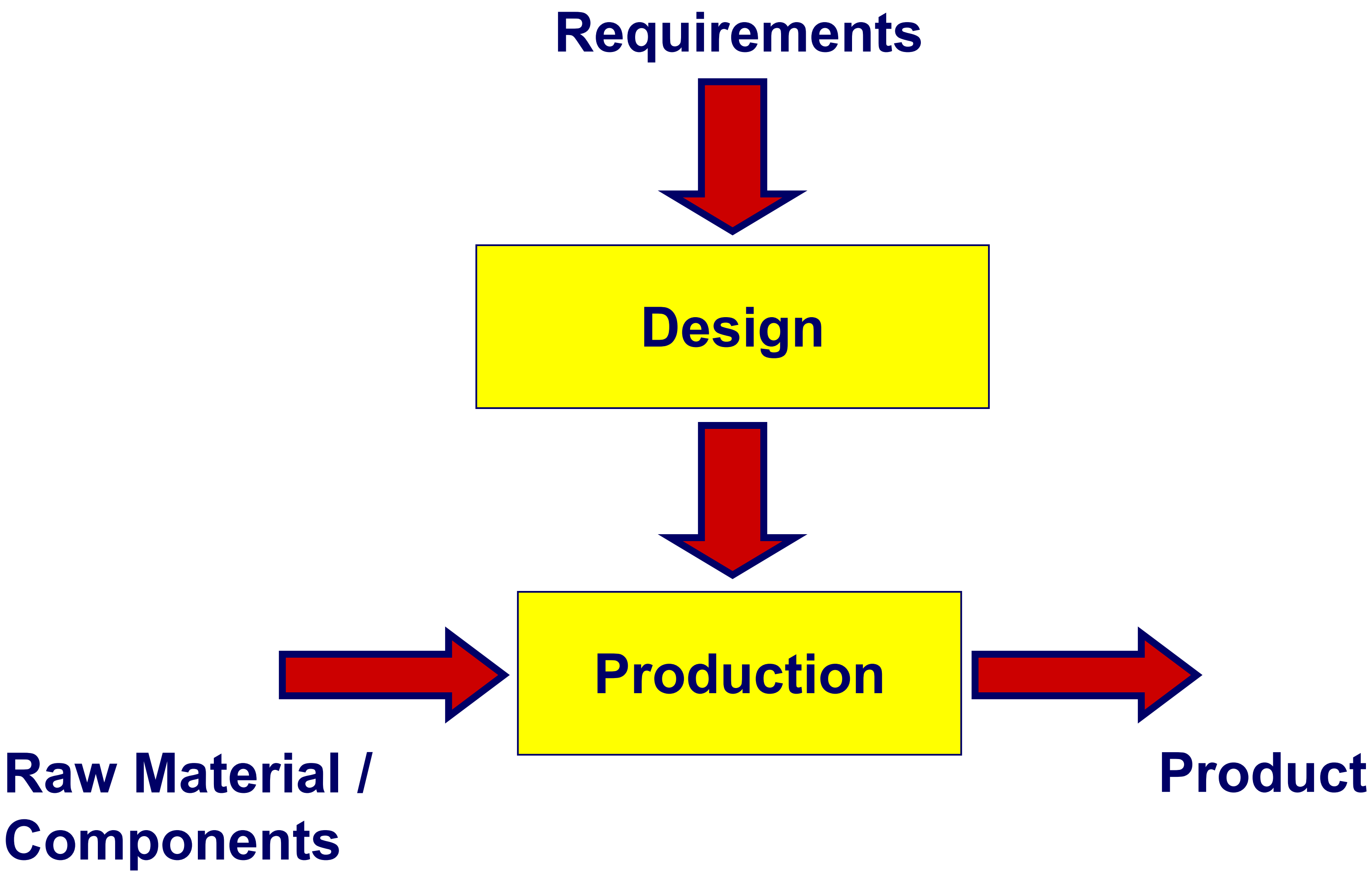
Introduction to Manufacturing

Basic Manufacturing Concepts


Requirements
Design
Production
Raw Material /
Components
Product

TYPES OF MANUFACTURING

Continuous

The elements composing the final product cannot be easily identified.

The process is not reversible.

Examples: Production of Iron, Paper, Chemicals, Pharmaceuticals, Food, …

Discrete

Finite number of components.

Two main phases:

- **Fabrication phase**: set of processes for modifying shape, dimensions, and surface state of a single part.
- **Assembly phase**: set of processes for joining single parts to obtain a (assembled) product.

Examples: Production of Cars, Computers, Appliances, Shoes, Toys, Phones, …

DISCRETE MANUFACTURING

Juicy Salif

Alessi

Design by Philippe Starck

Discrete Manufacturing

Tolomeo

Artemide

Design by Michele De Lucchi

DISCRETE MANUFACTURING

AW101

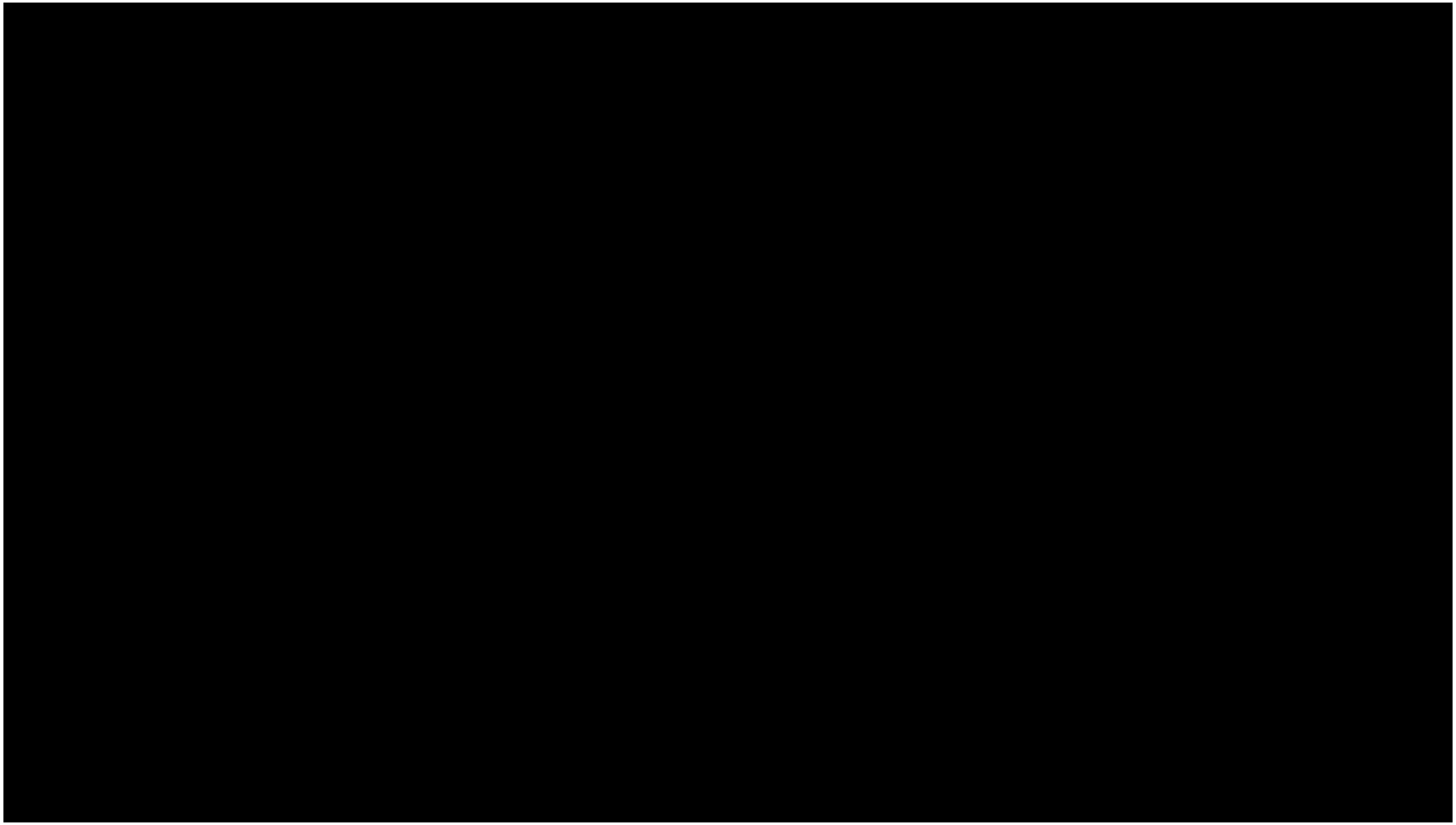
Leonardo

Helicopter

Discrete Manufacturing

Clock EVO
Machining Center
MCM spa

DISCRETE MANUFACTURING



https://youtu.be/lJx6cF-H__I

A Simplified Model of Manufacturing Process

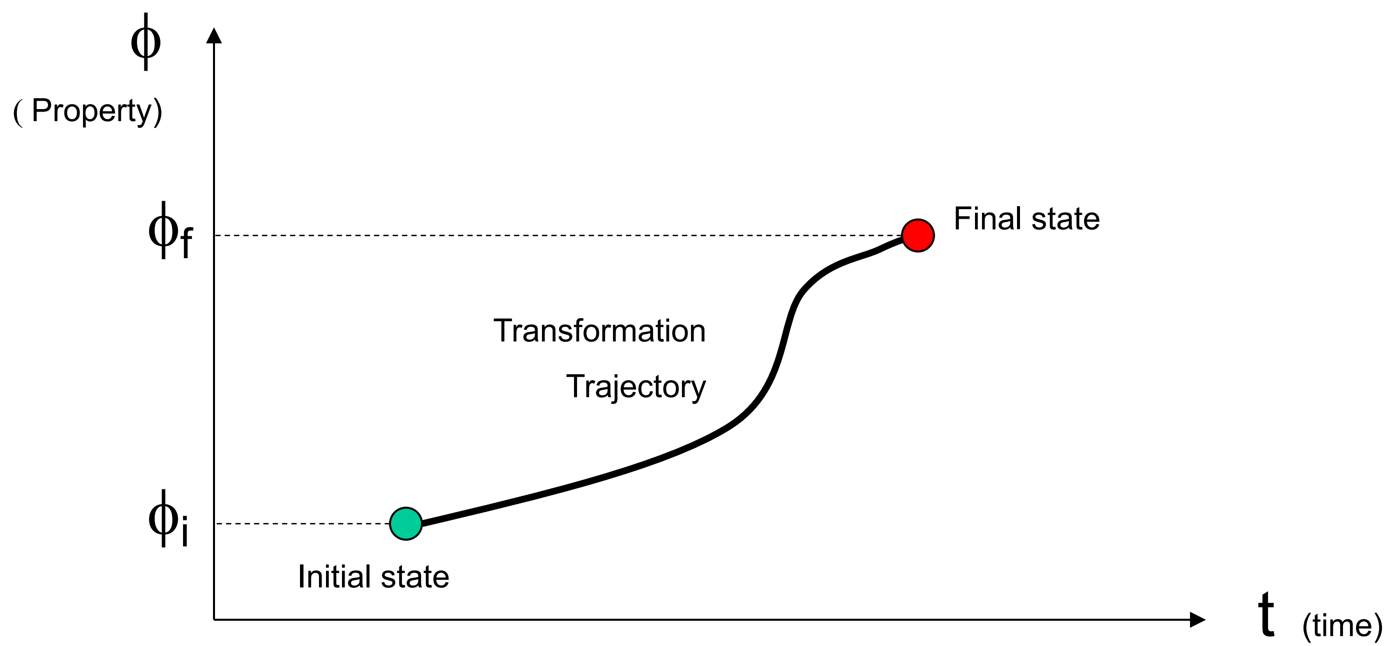
“The manufacturing process begins when raw materials are transported to a place – a factory – where labor, know-how, technology, equipment and energy are used to physically and/or chemically **transform** these materials into a product.

This product is then transported either to another factory, where these same inputs are again applied, or to a place, where those who wish to own it (consumers or other businesses that use the product for their activities) can purchase it.”

A Framework for Revitalizing American Manufacturing
Barack Obama - December 2009

Transformation Process

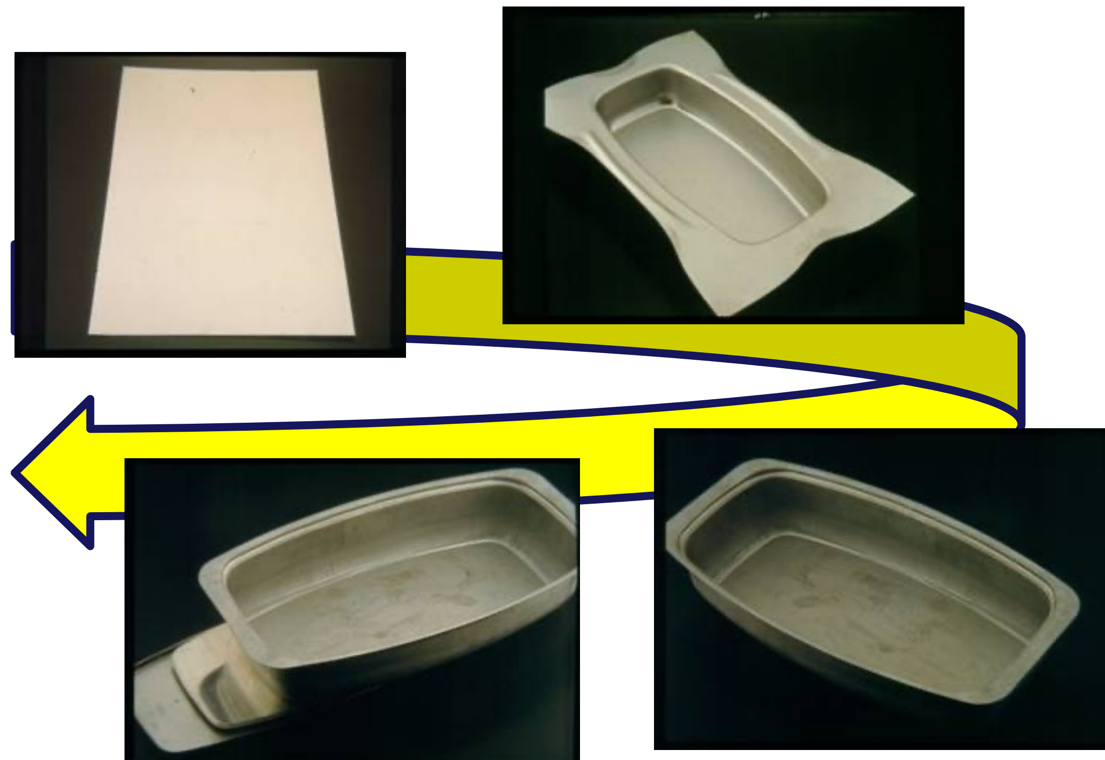
Modification of a set of product properties obtained by means of selected elementary processes.



RELEVANT TRANSFORMATION

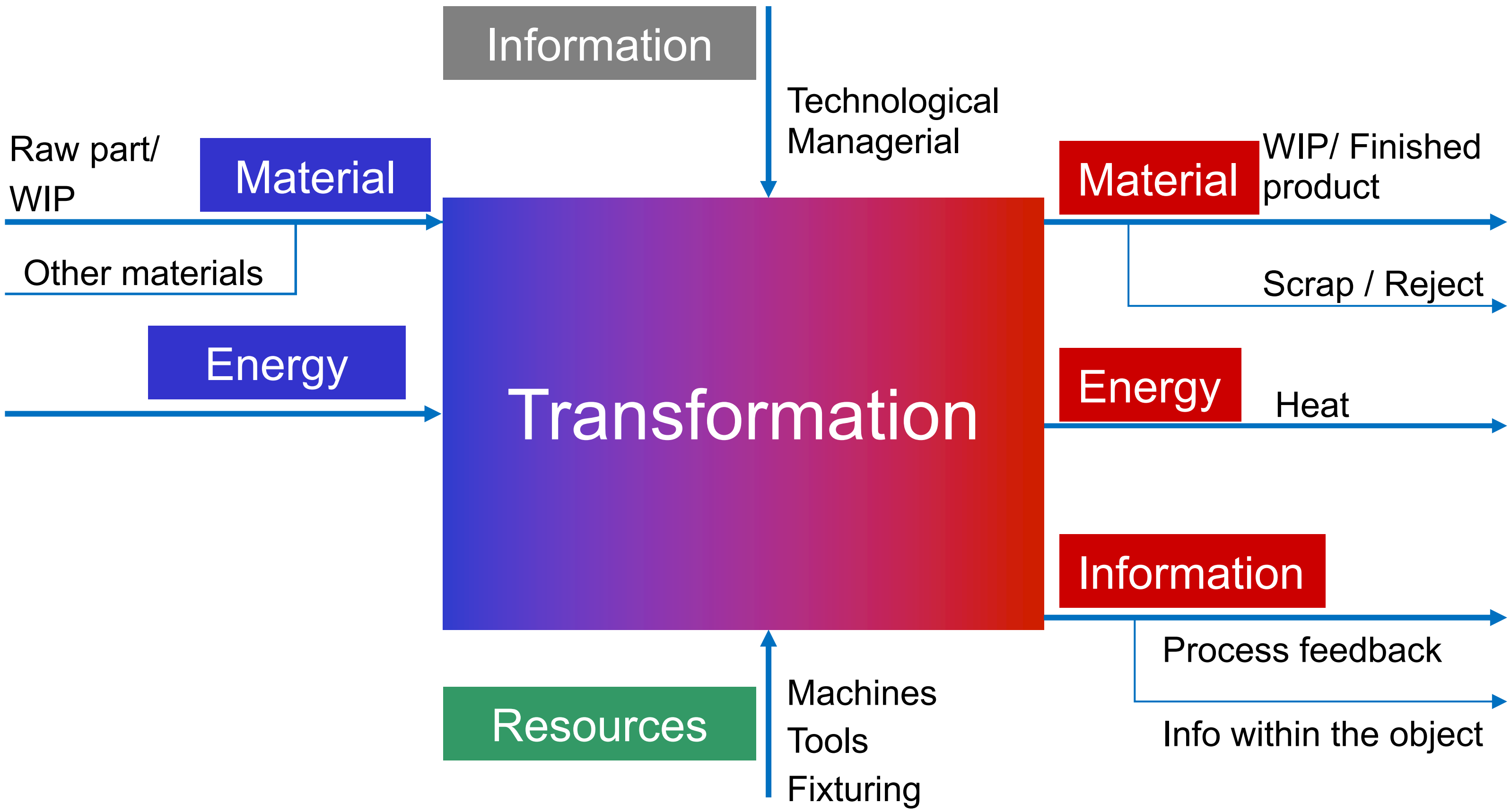
- Shape and dimension (macro-geometry)
- Surface finish (micro-geometry)
- Mechanical characteristics (stiffness, ductility, etc.)
- State, temperature

Fish Kettle
(stainless steel 304, thickness 1,2 mm)

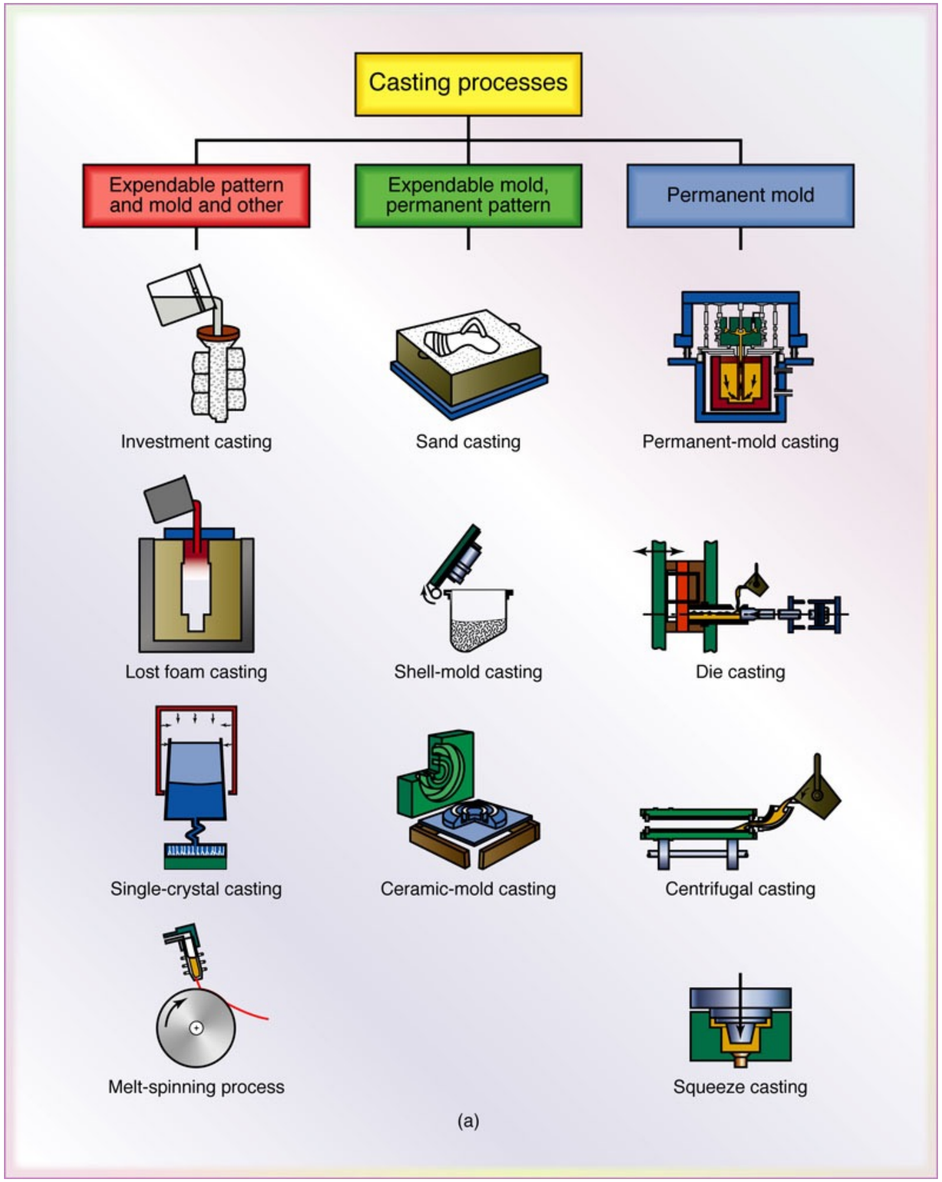


(courtesy of Serafino Zani, Lumezzane Gazzolo (BS))

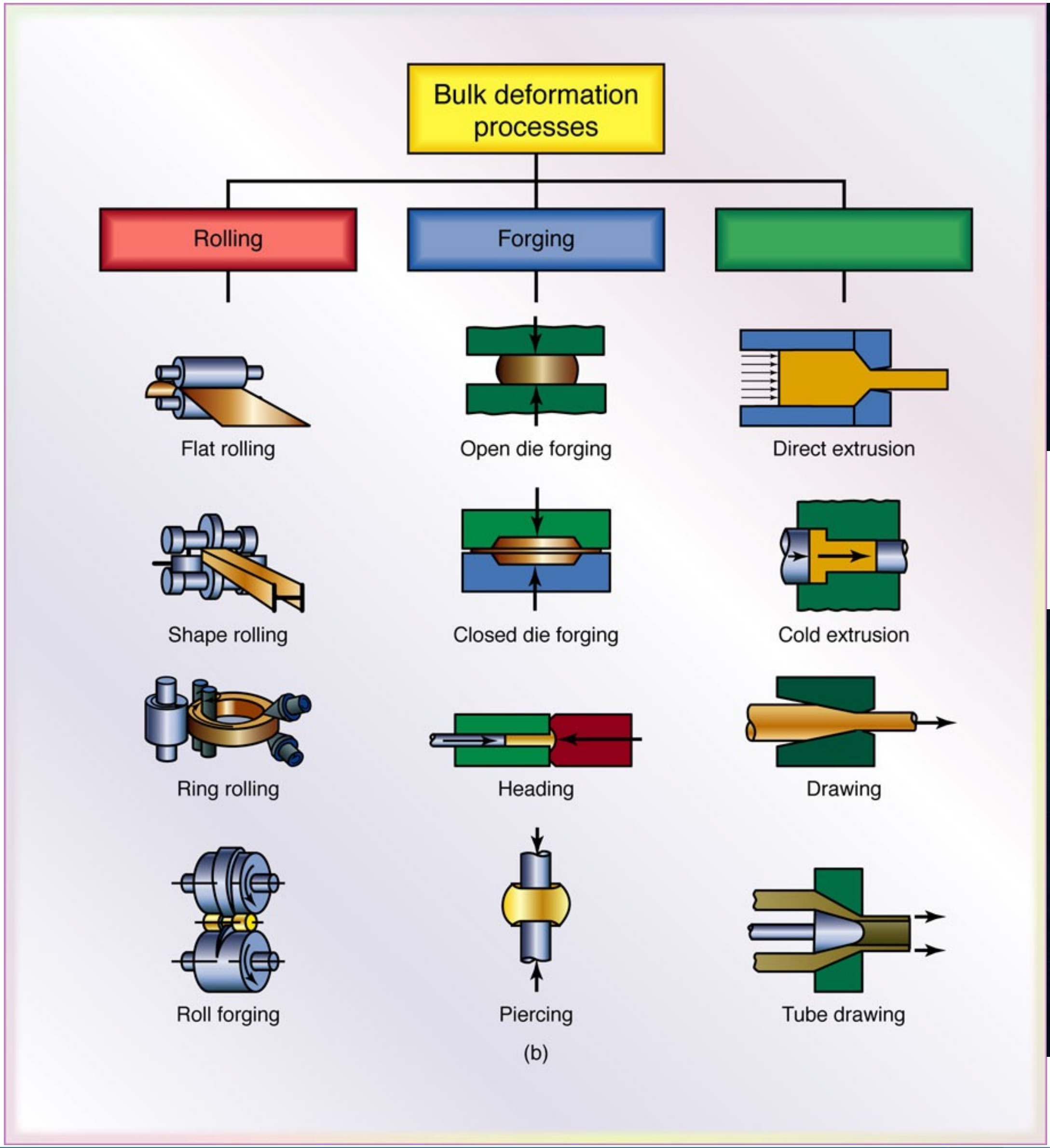
TRANSFORMATION - SCHEME

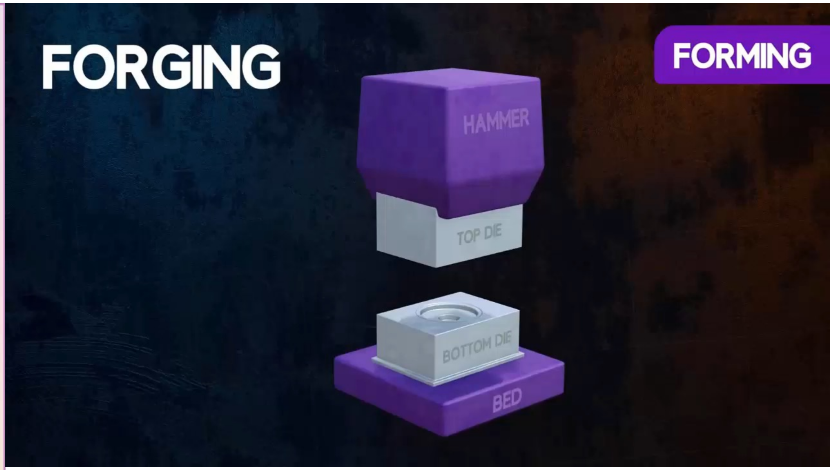


Methods of Transformations: Casting



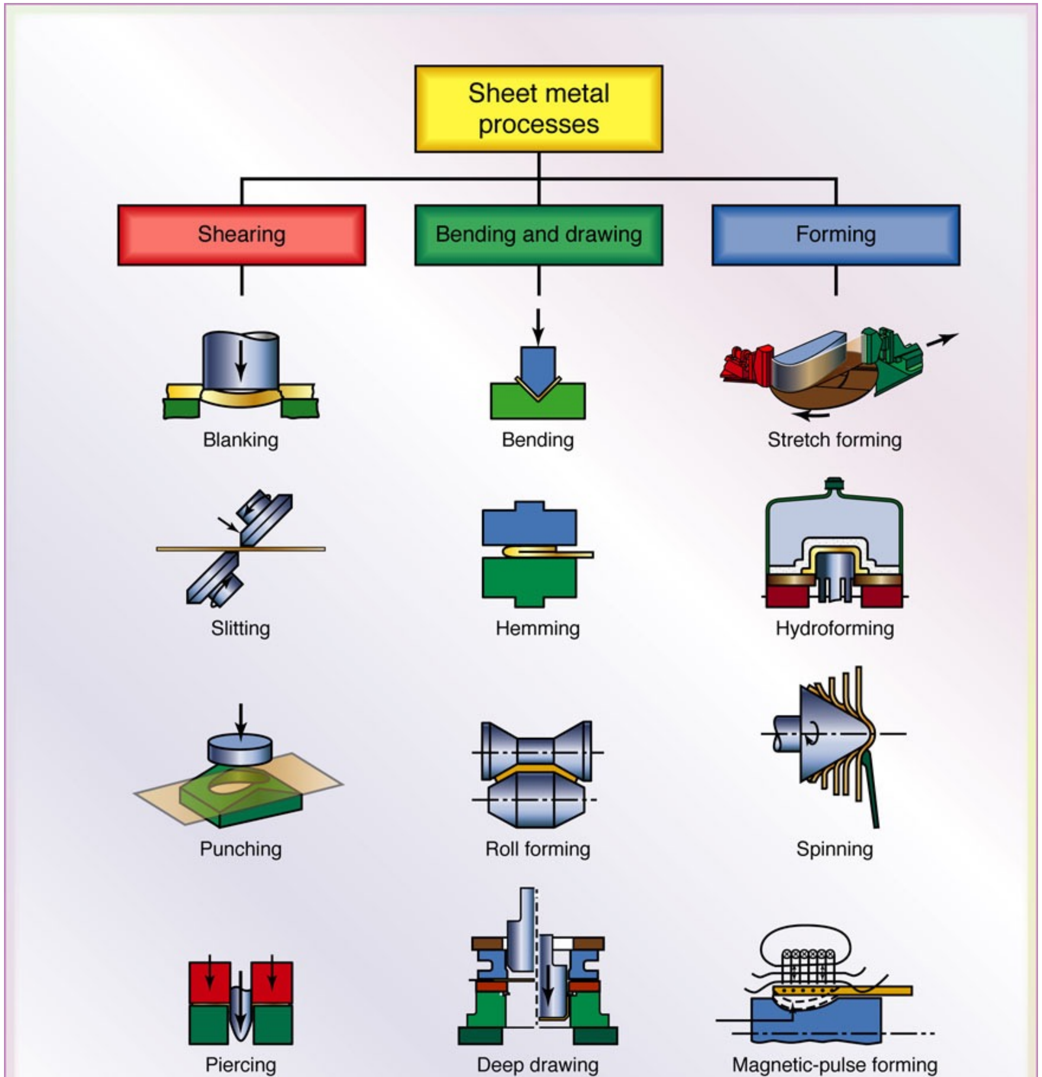
Methods of Transformations: Forming and Shaping





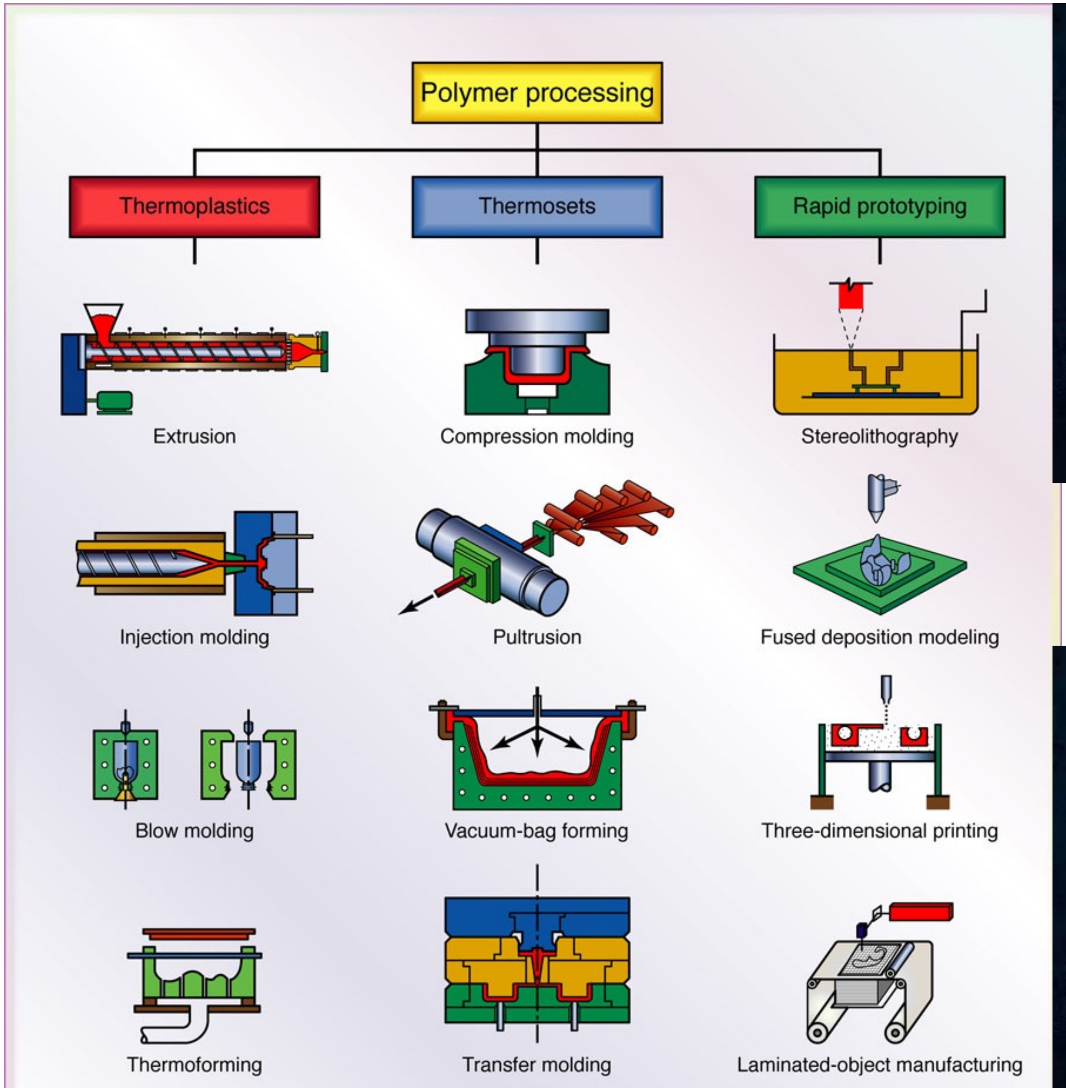
THE EFFICIENT ENGINEER

Methods of Transformations: Forming and Shaping



(c)

Methods of Transformations: Forming and Shaping



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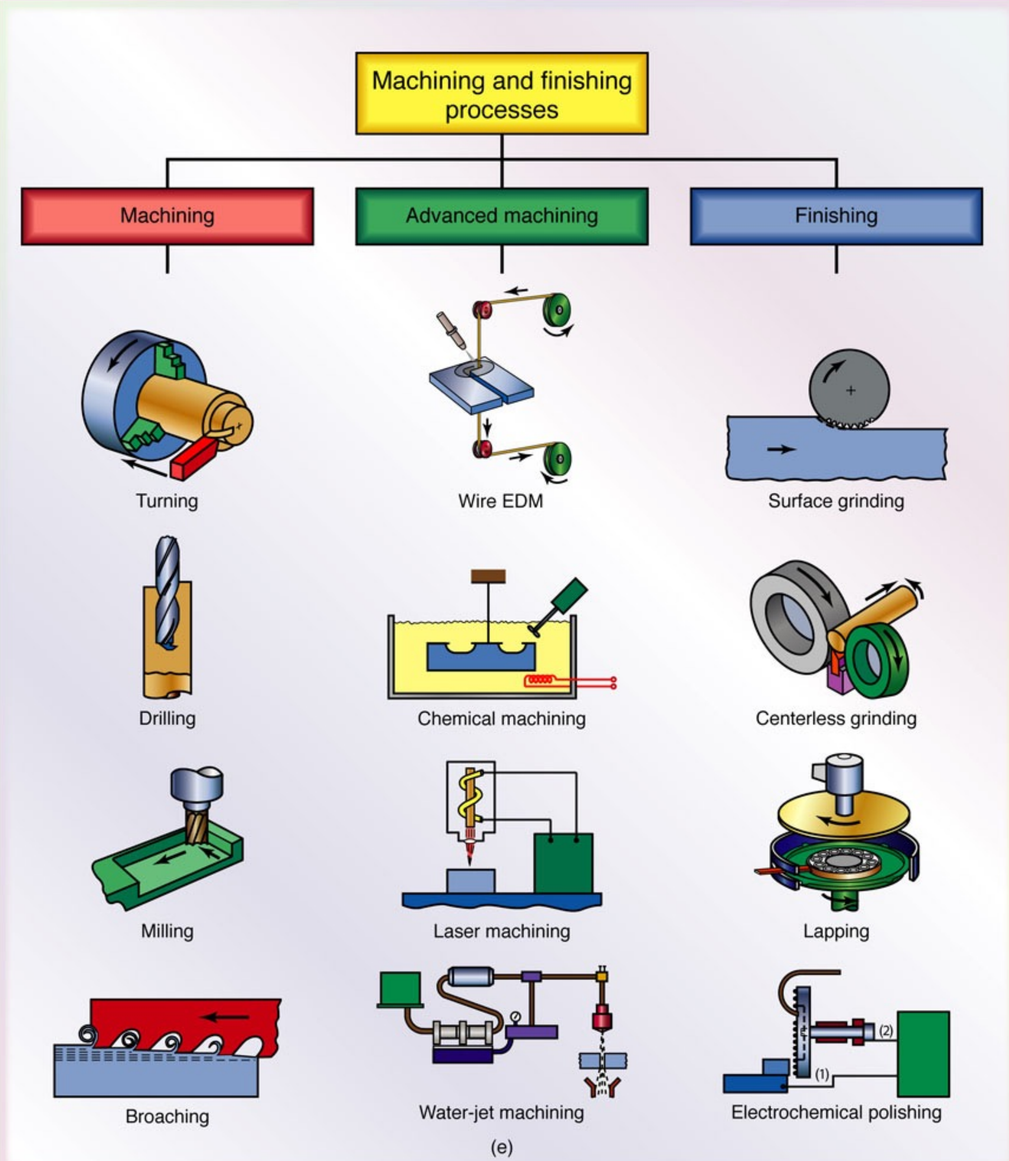
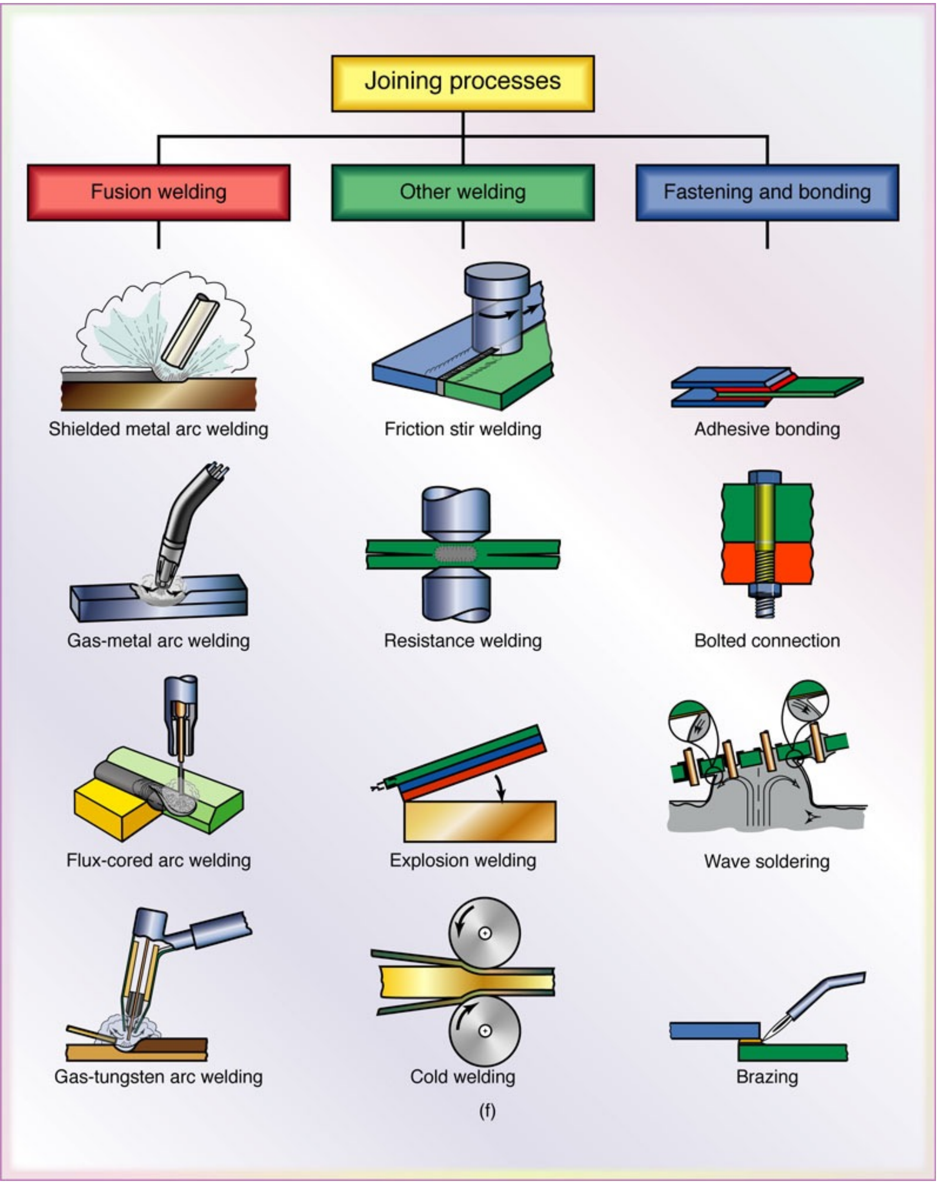
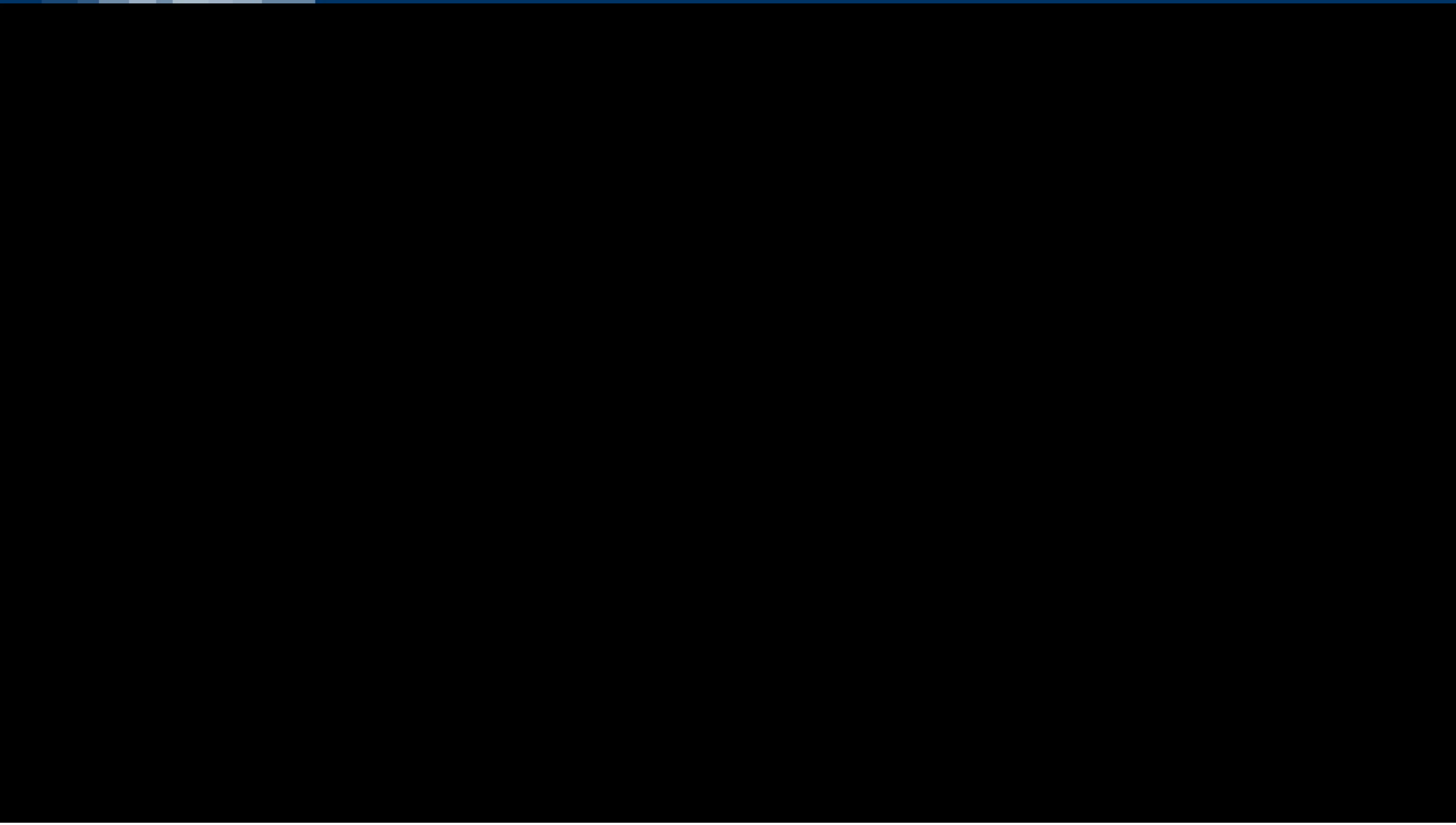
METHODS OF TRANSFORMATIONS: MACHINING



METHODS OF TRANSFORMATIONS: JOINING



EXAMPLE – ASSEMBLY



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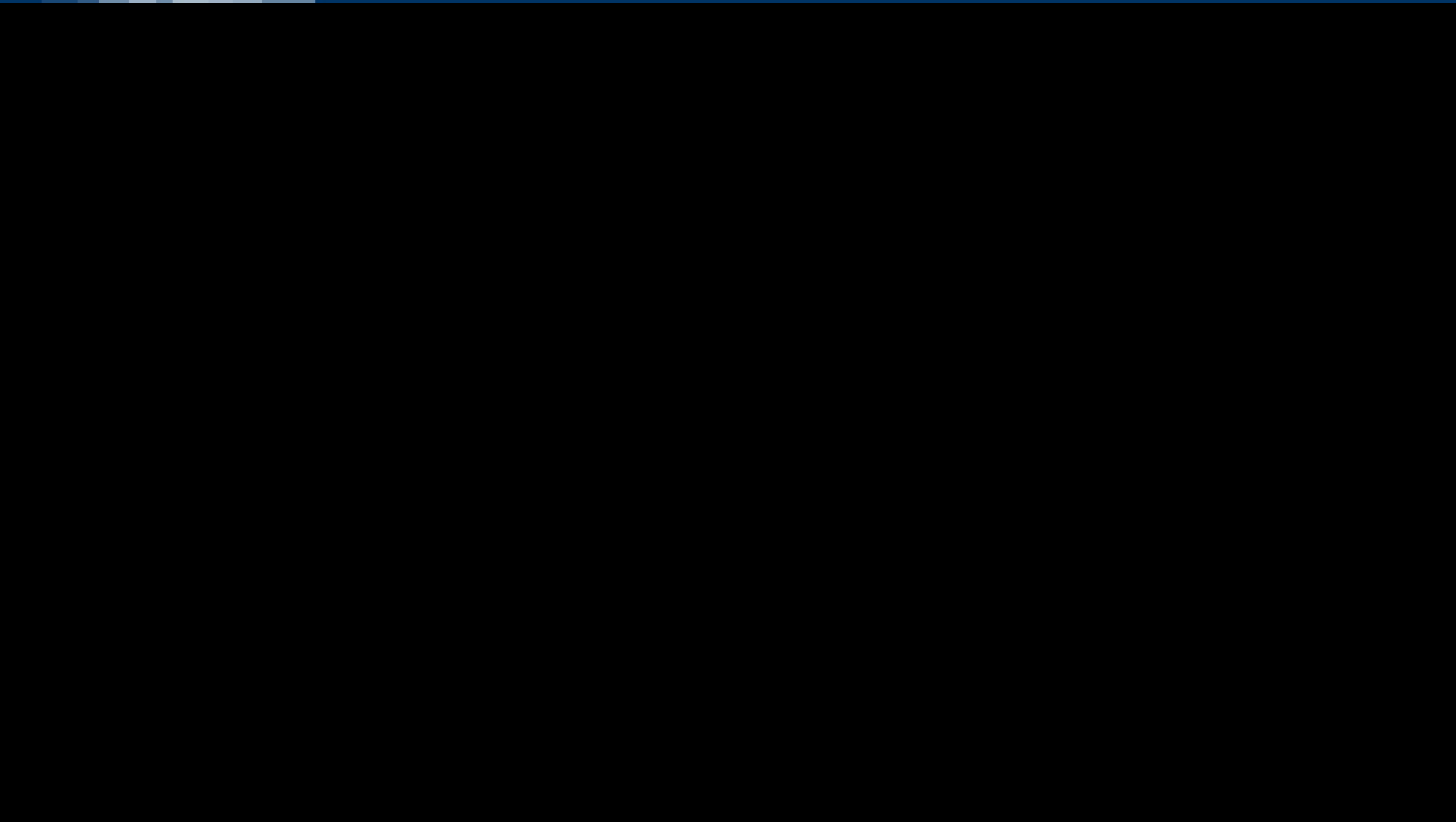
EXAMPLE – CHAIN OF TRANSFORMATIONS



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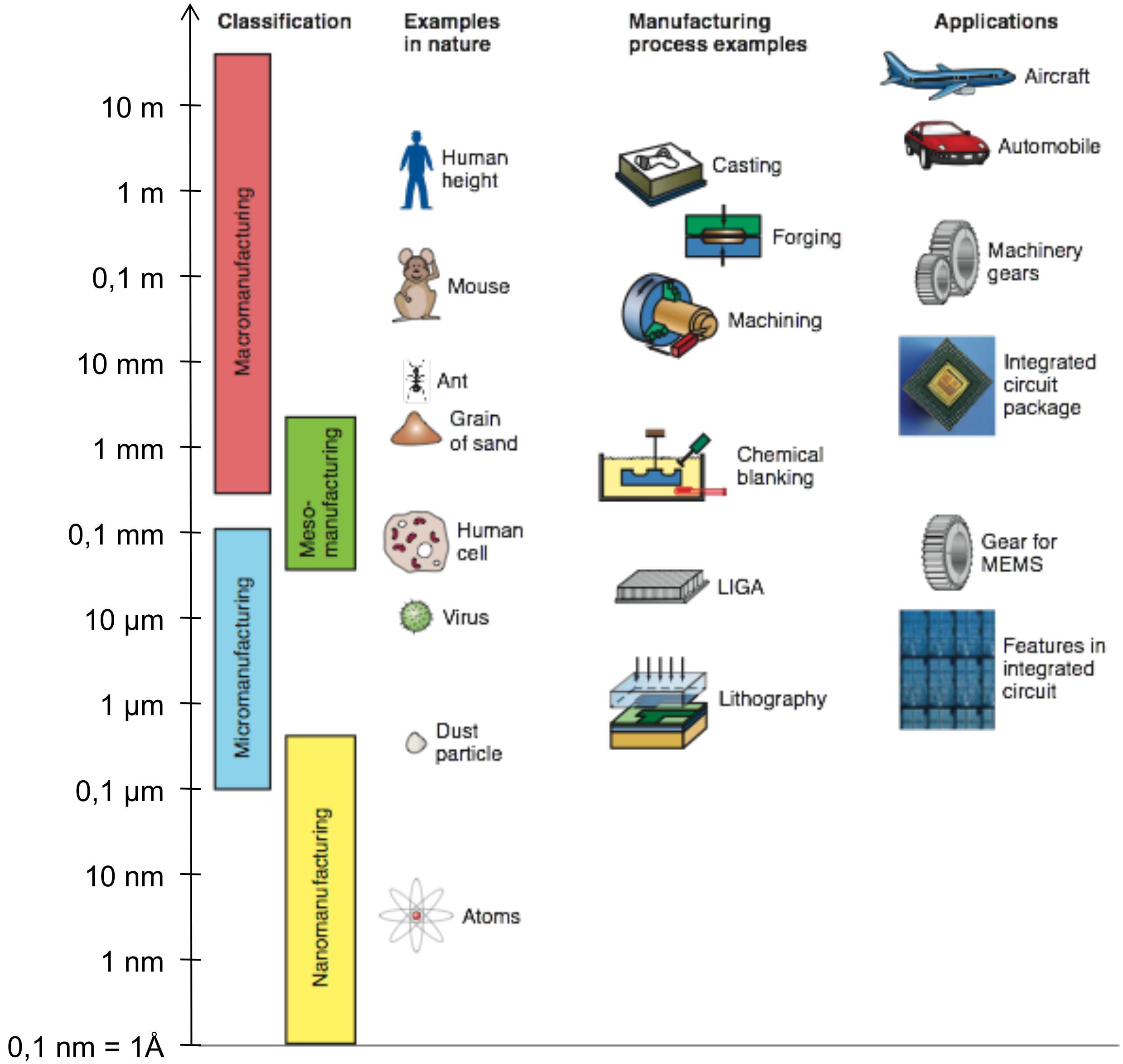
SCALES OF TRANSFORMATIONS



Technical Feasibility

Transformations (manufacturing processes)

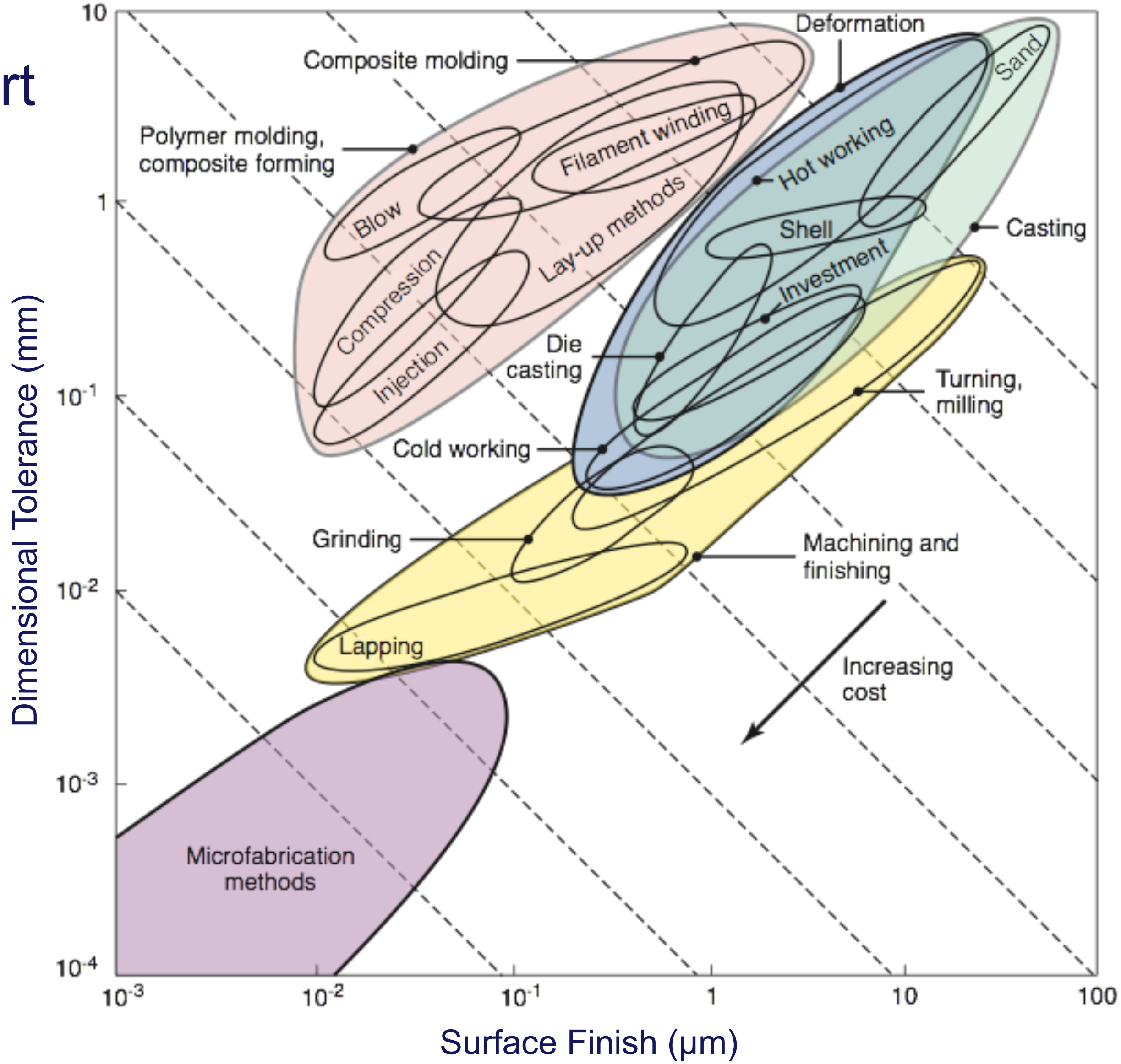
should guarantee:

- Technical specifications
- Production time
- Environmental regulations
- Operating safety

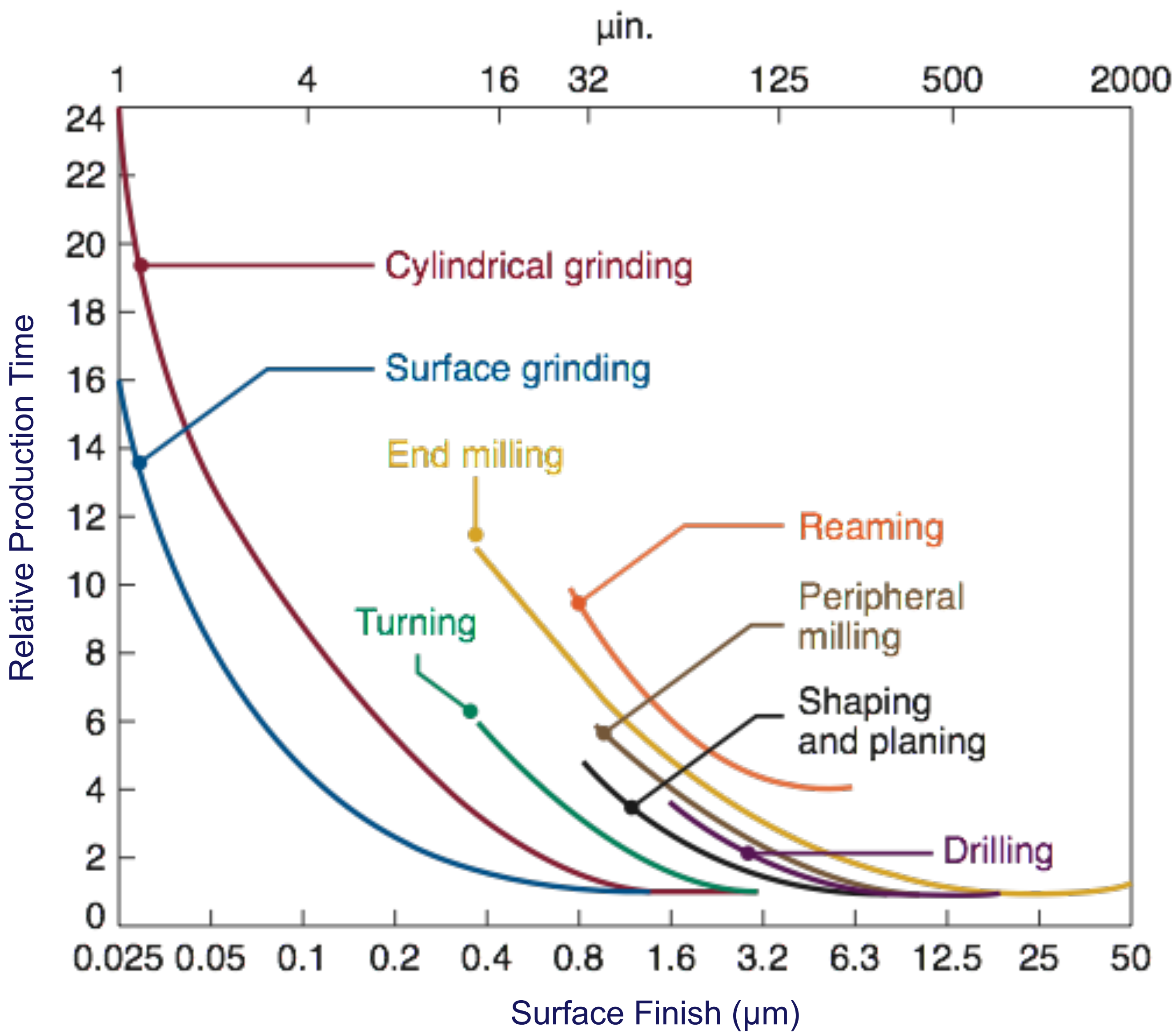
Feasibility and Sustainability

Technical Feasibility

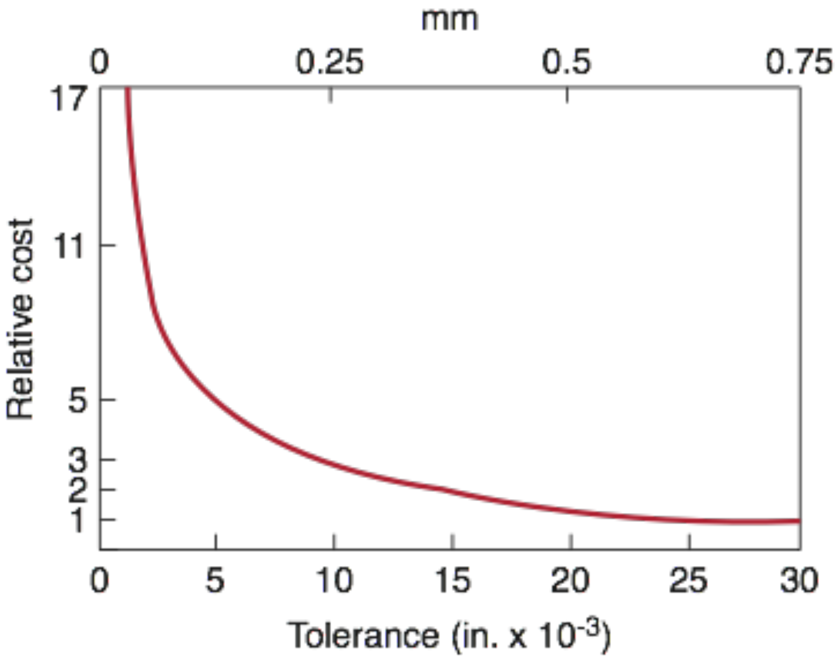
Ashby Chart



PRODUCTION TIME



Economical feasibility



Type of machinery	Price range ($000)	Type of machinery	Price range ($000)
Broaching	10-300	Machining center	50-1000
Drilling	10-100	Mechanical press	20-250
Electrical discharge	30-150	Milling	10-250
Electromagnetic	50-150	Ring rolling	>500
Extruder	30-80	Robot	20-200
Fused deposition modeling	40-200	Roll forming	5-100
Gear shaping	100-200	Rubber forming	50-500
Grinding		Stereolithography	80-500
Cylindrical	40-150	Stretch forming	400 - > 1000
Surface	20-100	Transfer line	100 - > 1000
Headers	100-150	Welding	
Injection molding	30-200	Electron beam	75-1000
Jig boring	50-150	Gas tungsten arc	1-5
Horizontal boring mill	100-400	Laser beam	60-1000
Flexible manufacturing system	> 1000	Resistance, spot	20-50
Lathe	10-100	Ultrasonic	50-200
Automatic	30-250		
Vertical turret	100-400		

Note: Prices vary significantly, depending on size, capacity, options, and level of automation and computer controls.

Choice of Manufacturing Process

Technological feasibility

Profitability

Manufacturing Process

LEARNING TARGETS